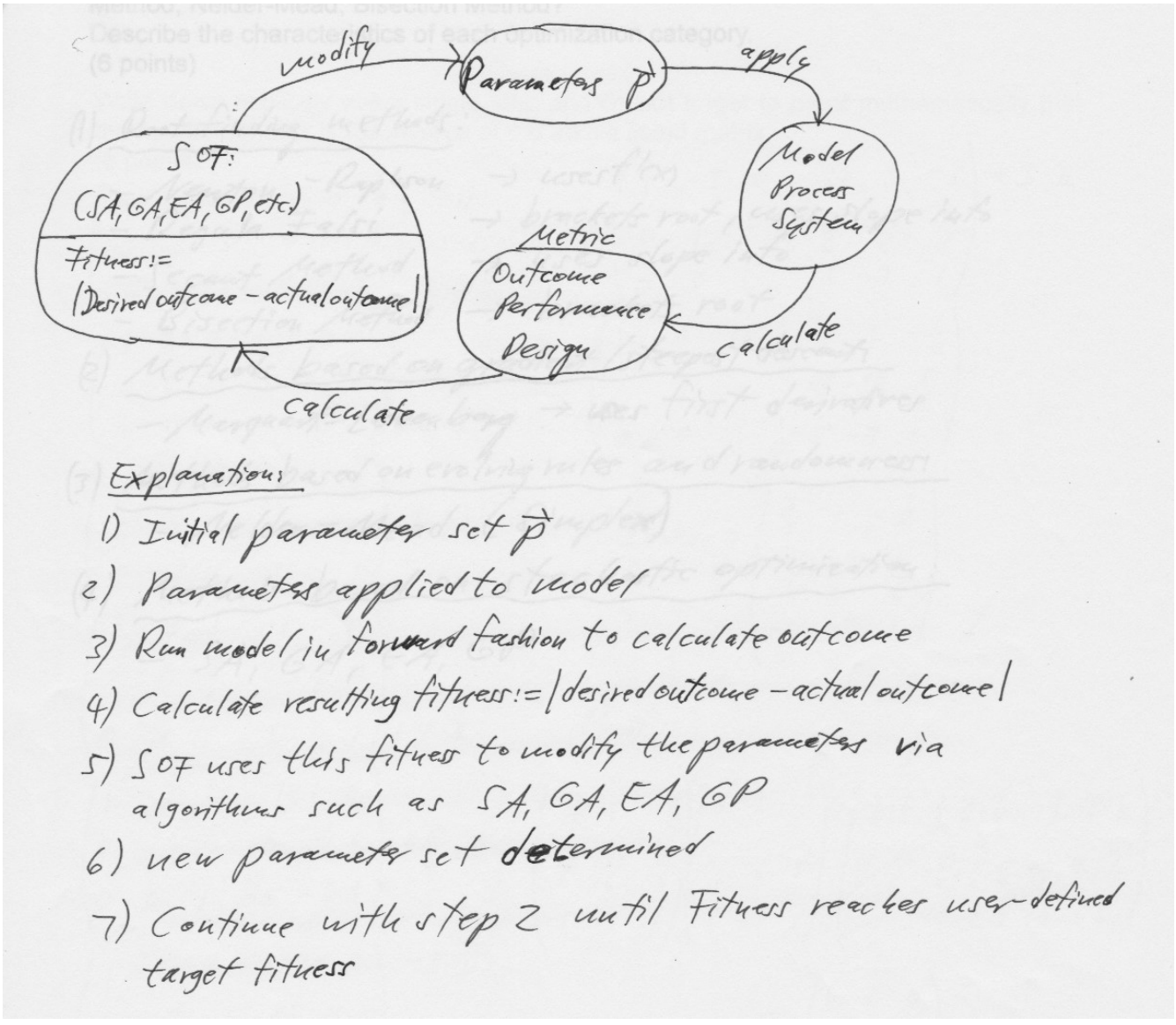
Questions for Final Exam ECE 330:

Name: _____

Question #1 (10 points total):

Describe the *Stochastic Optimization Framework* in detail (i.e., graphical sketch with labeling of all components and detailed written explanation of how it works).
(10 points)



Explanations:

1) Initial parameter set $\vec{p}$
2) Parameters applied to model
3) Run model in forward fashion to calculate outcome
4) Calculate resulting fitness:=|desired outcome - actual outcome|
5) SOF uses this fitness to modify the parameters via algorithms such as SA, GA, EA, GP
6) new parameter set determined
7) Continue with step 2 until Fitness reaches user-defined target fitness

Question #2 (10 points total):

Derive explicitly the update formula for the multi-dimensional Newton-Raphson Method.
(10 points)

have N functional relations to be zeroed with variables x_i with $i = 1, \dots, N$

$$F_i(x_1, x_2, \dots, x_N) = 0 \quad \forall i = 1, 2, \dots, N$$

$$\Rightarrow \vec{F}(\vec{x}) = 0$$

in neighborhood of $\vec{x}$ each function $F_i(\vec{x})$ can be expanded in Taylor series:

$$F_i(\vec{x} + \delta\vec{x}) = F_i(\vec{x}) + \underbrace{\sum_{j=1}^{N} \frac{\partial F_i(\vec{x})}{\partial x_j}}_{\text{matrix of partial derivatives of } F_i} \delta x_j + \mathcal{O}(\delta \vec{x}^2) \quad \forall i = 1, 2, \dots, N$$

is Jacobian matrix $\mathcal{J}$: $J_{ij} = \frac{\partial F_i(\vec{x})}{\partial x_j}$

$$\Rightarrow \vec{F}(\vec{x} + \delta\vec{x}) = \vec{F}(\vec{x}) + \mathcal{J}\,\delta\vec{x} + \cancel{\mathcal{O}(\delta\vec{x}^2)}$$

ignore error terms

and set $\vec{F}(\vec{x} + \delta\vec{x}) = 0$

$$\Rightarrow 0 = \vec{F}(\vec{x}) + \mathcal{J}\,\delta\vec{x}$$

$$\Rightarrow \mathcal{J}\,\delta\vec{x} = -\vec{F}(\vec{x})$$

$$\Rightarrow \delta\vec{x} = -\mathcal{J}^{-1}\vec{F}(\vec{x})$$

$\Rightarrow$ Newton-Raphson Method in N Dimensions

N update equations:

$$\vec{x}_{new} = \vec{x}_{old} + \delta\vec{x}$$

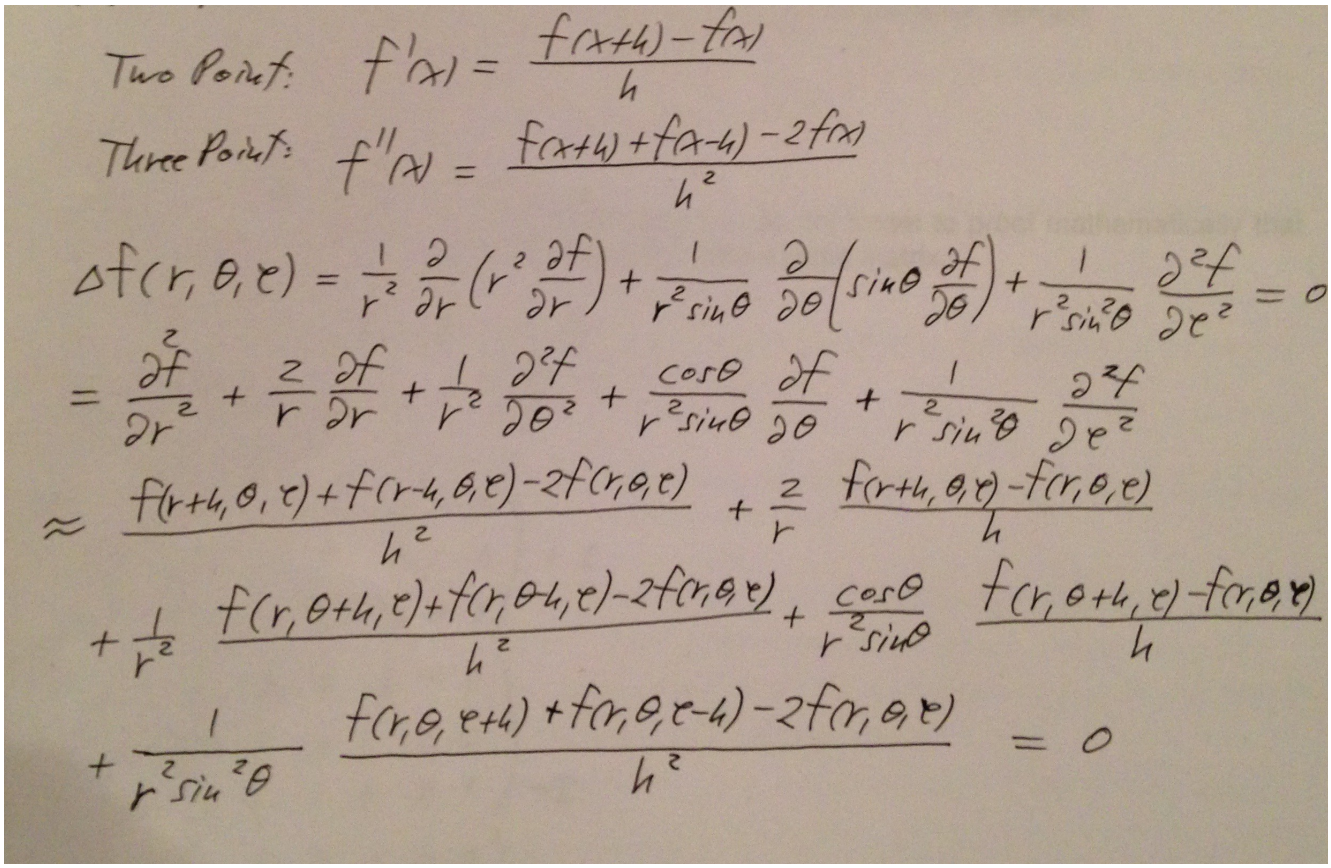
Question #3 (12 points total):

Approximate the general Laplace equation in spherical coordinates with the Two-Point and Three-Point Formulas and ignore the respective leading error terms. Hint: you have to differentiate the brackets first before applying the approximation formulas.

The Laplace equation in spherical coordinates is as follows:

$$\Delta f = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial f}{\partial \theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial \varphi^2}$$

(12 points)



Two Point: $f'(x) = \frac{f(x+h)-f(x)}{h}$

Three Point: $f''(x) = \frac{f(x+h)+f(x-h)-2f(x)}{h^2}$

$$\Delta f(r,\theta,\varphi) = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial f}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial f}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\varphi^2} = 0$$

$$= \frac{\partial^2 f}{\partial r^2} + \frac{2}{r}\frac{\partial f}{\partial r} + \frac{1}{r^2}\frac{\partial^2 f}{\partial\theta^2} + \frac{\cos\theta}{r^2\sin\theta}\frac{\partial f}{\partial\theta} + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 f}{\partial\varphi^2}$$

$$\approx \frac{f(r+h,\theta,\varphi)+f(r-h,\theta,\varphi)-2f(r,\theta,\varphi)}{h^2} + \frac{2}{r}\frac{f(r+h,\theta,\varphi)-f(r,\theta,\varphi)}{h}$$

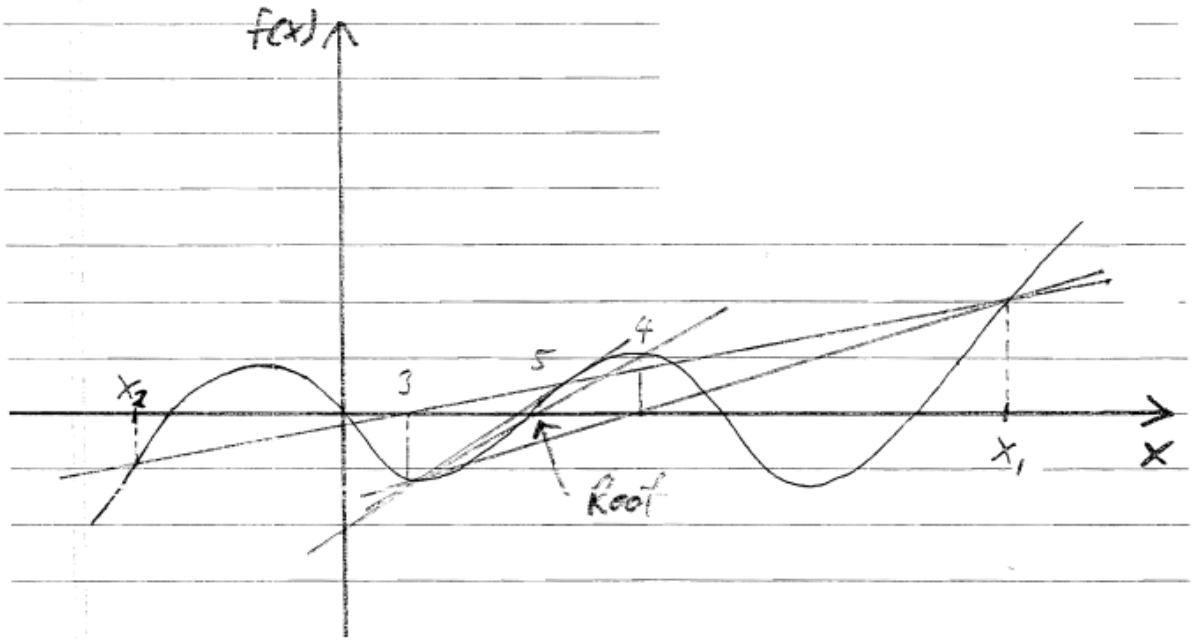
$$+ \frac{1}{r^2}\frac{f(r,\theta+h,\varphi)+f(r,\theta-h,\varphi)-2f(r,\theta,\varphi)}{h^2} + \frac{\cos\theta}{r^2\sin\theta}\frac{f(r,\theta+h,\varphi)-f(r,\theta,\varphi)}{h}$$

$$+ \frac{1}{r^2\sin^2\theta}\frac{f(r,\theta,\varphi+h)+f(r,\theta,\varphi-h)-2f(r,\theta,\varphi)}{h^2} = 0$$

Question #4 (6 points total):

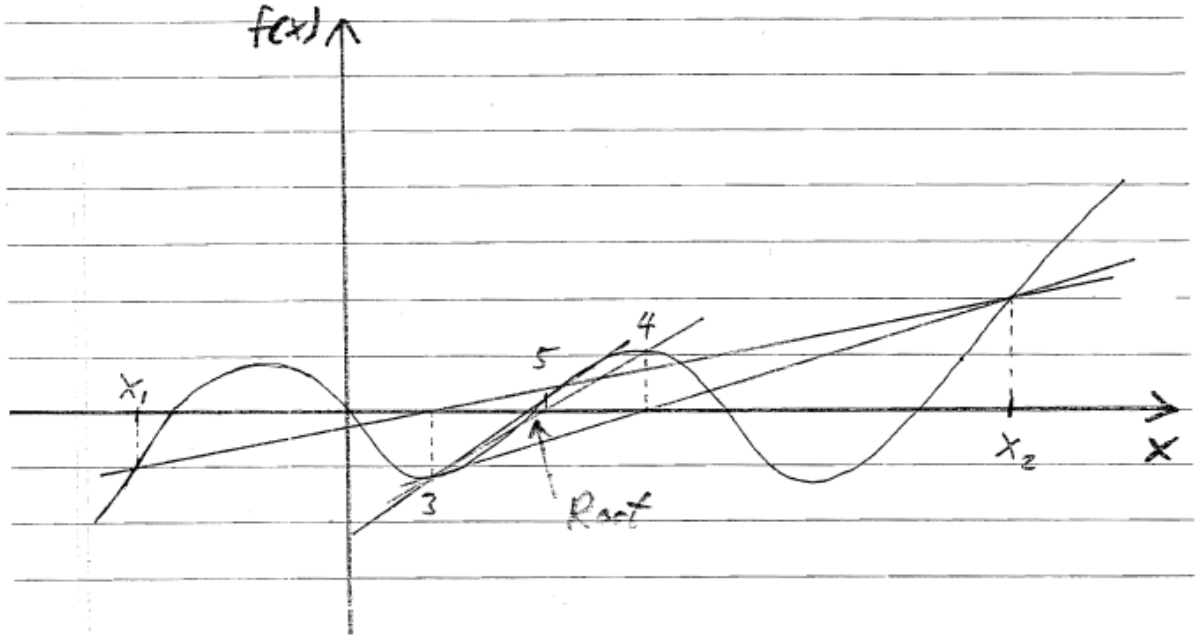
a) Applying the Regula Falsi with x_1 and x_2 as the two starting x-values, which root of the given graph will you find? Finding the correct result will require you to be extremely precise when executing the method (use of a ruler highly recommended!). Draw as many iterations as necessary to demonstrate convincingly which root will be found. NO GUESSING!

(3 points)



b) Which root will you find if x_1 and x_2 are reversed? Explain the respective results from a and b. → **Explanation:** same result as in a) because Regula Falsi brackets root at all times.

(3 points)



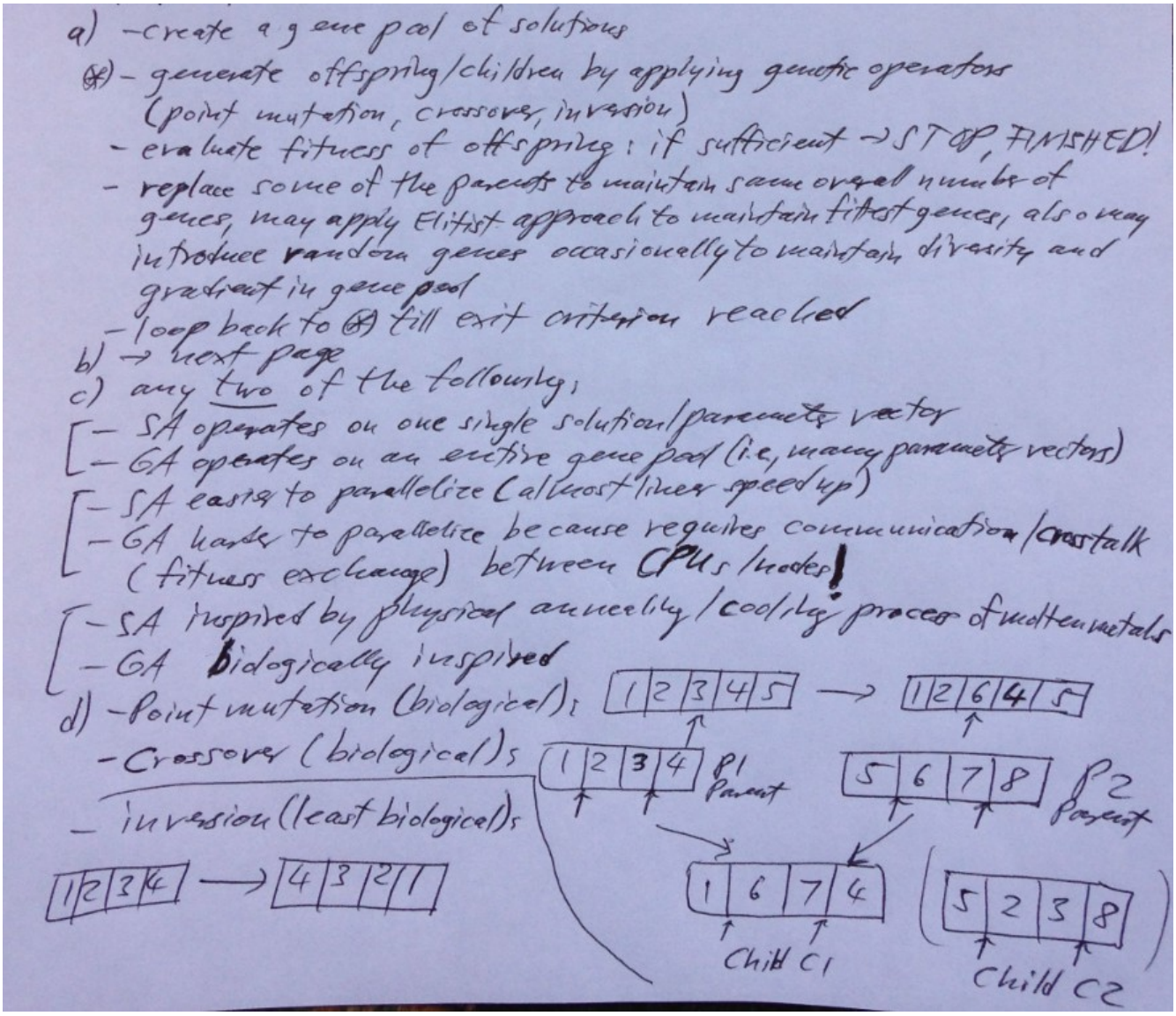
Question #5 (28 points total):

a) Describe in detailed words the general procedure for Genetic Algorithms (GA).
(6 points)

b) Describe the Simulated Annealing (SA) algorithm in detailed pseudo-code form.
(10 points)

c) What are the main differences between GA and SA?
(2 points)

d) List three genetic operators by name, illustrate with concrete examples how they work, and point out the ones that are biologically motivated.
(10 points)



b) detailed SA pseudo-code:

```
Set Temperature T = T0
Initial parameter vector P
Calculate initial current fitness/energy E_current
Set best fitness/energy E_best = E_current
Set best parameter vector P_best = P
Set cooling rate: 0 < lambda < 1

LOOP:
{
    Modify parameter vector P to yield P'
    Calculate resulting temporary fitness/energy: E_temp

    if (E_temp< E_current)
    {
        E_current = E_temp
        make P' the new P: P = P'
        if (E_current < E_best)
        {
            E_best = E_current
            P_best = P
            if (E_best < user-defined fitness/energy)
            {
                STOP optimization
            }
        }
    }
    else
    {
        if (random number [0...1] < exp[-(E_temp - E_best)/T])      [COMMENT: accept unfavorable step with Boltzmann-probability [0...1]]
        {
            E_current = E_temp
            make P' the new P: P = P'
        }
        else
        {
            reject P' and go back to P
        }
    }
    T = lambda * T      [COMMENT: cool temperature for Boltzmann-probability, i.e., make it harder to accept unfavorable steps]
    if (T < user-defined minimum T)
    {
        STOP optimization
    }
    else
    {
        goto LOOP
    }
}
```

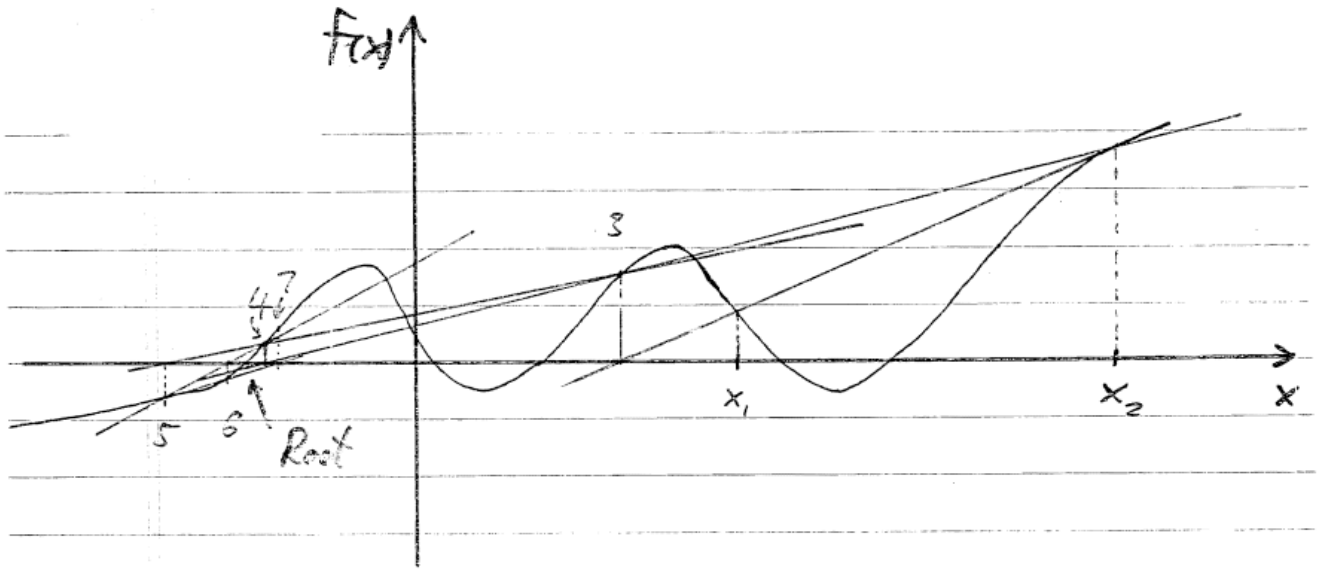
} exit criterion (I)

} exit criterion (II)

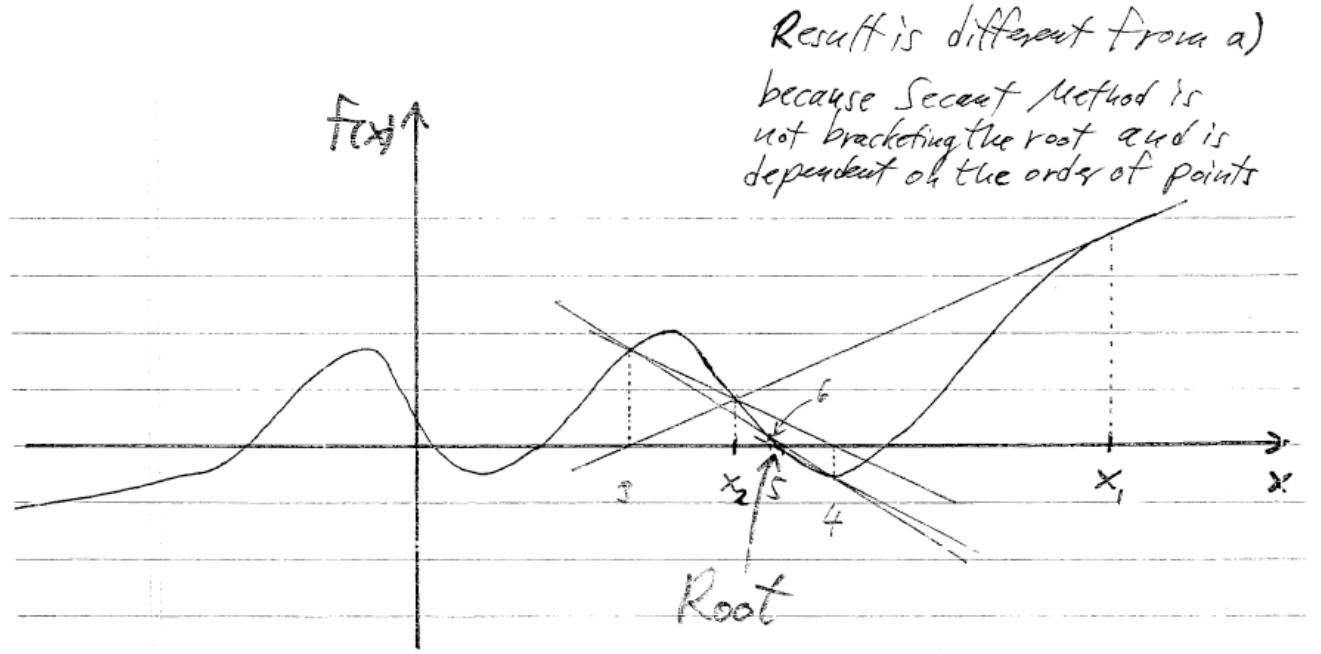
can also be max. number of allowed/prescribed iterations exhausted/exceeded

Question #6 (6 points total):

a) Applying the Secant Method with x_1 and x_2 as the two starting x-values, which root of the given graph will you find? Finding the correct result will require you to be extremely precise when executing the method (use of a ruler highly recommended!). Draw as many iterations as necessary to demonstrate convincingly which root will be found. NO GUESSING!
(3 points)



b) Which root will you find if x_1 and x_2 are reversed? Explain the respective results from a and b.
(3 points)



Question #7 (10 points total):

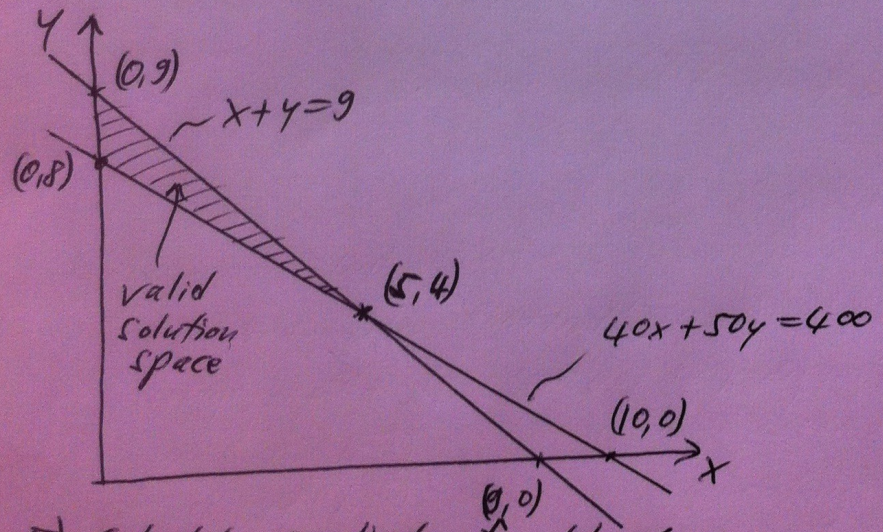
A company is preparing a trip for 400 employees. The transportation service has 10 buses of 50 seats each and 8 buses of 40 seats, but only 9 drivers available. The rental cost for a large bus is $800 and $600 for the small bus. Calculate how many buses of each type should be used for the trip for the least possible cost.
(10 points)

1) define unknowns:
x = # small buses
y = # big buses

2) Write objective function:
$f(x,y) = 600x + 800y$

3) Write constraints as system of inequalities:
$40x + 50y \geq 400$
$x + y \leq 9$
$x \geq 0$ $\quad x \in \mathbb{N}_0$ (integers)
$y \geq 0$ $\quad y \in \mathbb{N}_0$ (integers)

4) Find set of feasible solutions:



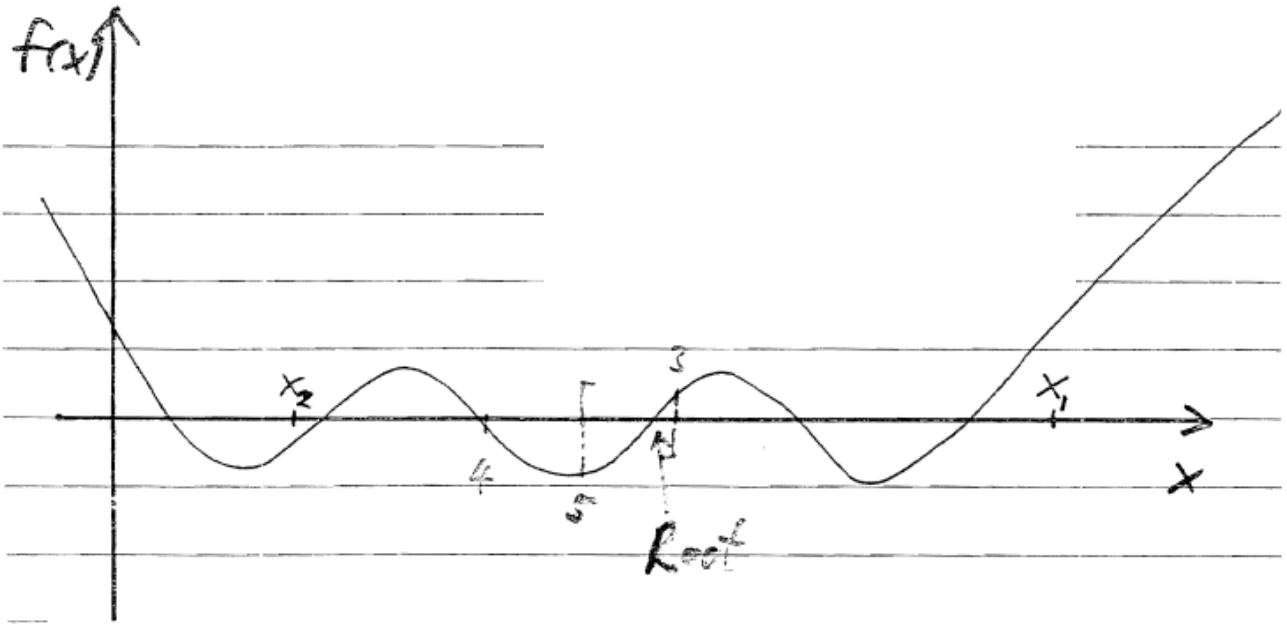
5) Calculate coordinates of valid vertices:
(I): $x+y=9 \Rightarrow (0,9)$
(II): $40x+50y=400 \Rightarrow (0,8)$
(III): $\left.\begin{matrix} x+y=9 \\ 40x+50y=400 \end{matrix}\right\} \Rightarrow (5,4)$

6) Calculate objective function value at each vertex:
$f(0,9) = 7200$
$f(0,8) = 6400$
$\boxed{f(5,4) = 6200}$

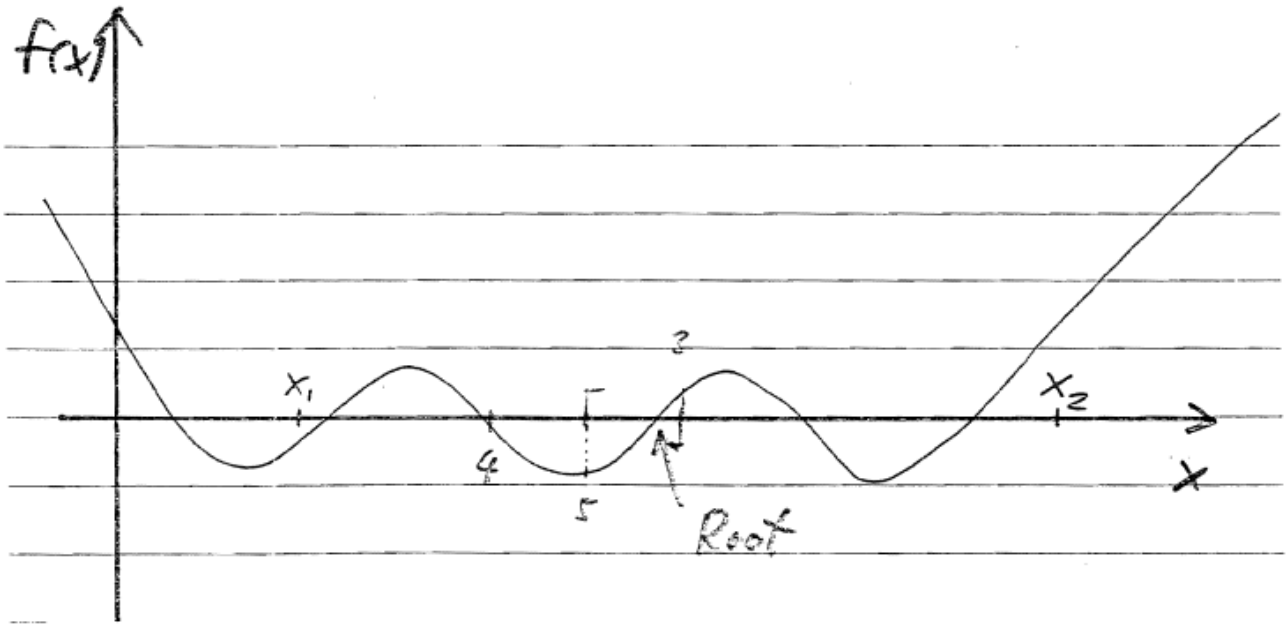
$\Rightarrow$ minimum cost achieved with 5 small and 4 large buses

Question #8 (6 points total):

a) Applying the Bisection Method with x_1 and x_2 as the two starting x-values, which root of the given graph will you find? Finding the correct result will require you to be extremely precise when executing the method (use of a ruler highly recommended!). Draw as many iterations as necessary to demonstrate convincingly which root will be found. NO GUESSING!
(3 points)



b) Which root will you find if x_1 and x_2 are reversed? Explain the result from a and b.
(3 points) → **<u>Explanation:</u>** same result as in a) because Bisection Method brackets root at all times.

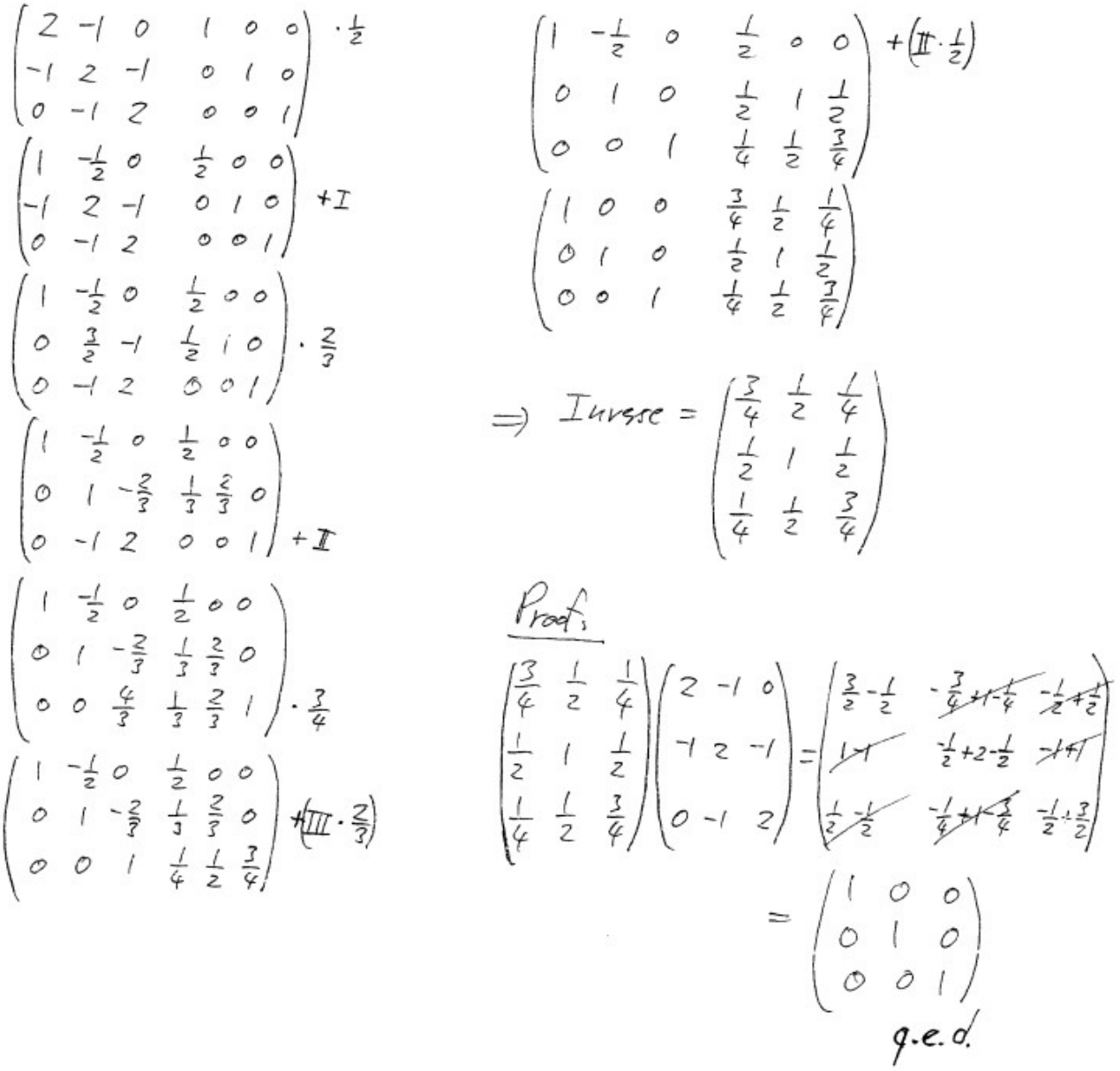


Question #9 (10 points total):

Invert the following matrix using the Gauss-Jordan-Elimination Method:

$$\begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix}$$

Write down explicitly every single step and do not forget to proof mathematically that the resulting matrix is the inverse of the above listed matrix.
(10 points)



$$\Rightarrow \text{Inverse} = \begin{pmatrix} \frac{3}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{pmatrix}$$

Proof:

$$\begin{pmatrix} \frac{3}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 1 & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{pmatrix} \begin{pmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{3}{2}-\frac{1}{2} & -\frac{3}{4}+1-\frac{1}{4} & -\frac{1}{2}+\frac{1}{2} \\ 1-1 & -\frac{1}{2}+2-\frac{1}{2} & -1+1 \\ \frac{1}{2}-\frac{1}{2} & -\frac{1}{4}+1-\frac{3}{4} & -\frac{1}{2}+\frac{3}{2} \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

q.e.d.

Question #10 (6 points total):

a) You have the following parameter vector in the population of a Genetic Algorithm:
$[1 \quad 3 \quad -1 \quad 9 \quad 4 \quad -5 \quad -7 \quad 8]$
Apply the *inversion* operator and write down the resulting parameter vector.
(2 points)

b) The following two parents have been selected for a *crossover* operation in a Genetic Algorithm:

Parent 1: $[1 \quad 3 \quad -1 \quad 9 \quad 4 \quad -5 \quad -7 \quad 8]$

Parent 2: $[8 \quad -2 \quad 0 \quad -1 \quad 6 \quad 3 \quad 9 \quad 0]$

The crossover area starts at position 4 (including) and ends at position 7 (including). Which two resulting offspring parameter vectors will result?
(4 points)

a) $[8 \quad -7 \quad -5 \quad 4 \quad 9 \quad -1 \quad 3 \quad 1]$

b) Parent 1: $[1 \quad 3 \quad -1 \quad 9 \quad 4 \quad -5 \quad -7 \quad 8]$
[----------]

Parent 2: $[8 \quad -2 \quad 0 \quad -1 \quad 6 \quad 3 \quad 9 \quad 0]$
[----------]

Child 1: $[1 \quad 3 \quad -1 \quad -1 \quad 6 \quad 3 \quad 9 \quad 8]$

Child 2: $[8 \quad -2 \quad 0 \quad 9 \quad 4 \quad -5 \quad -7 \quad 0]$

Question #11 (10 points total):

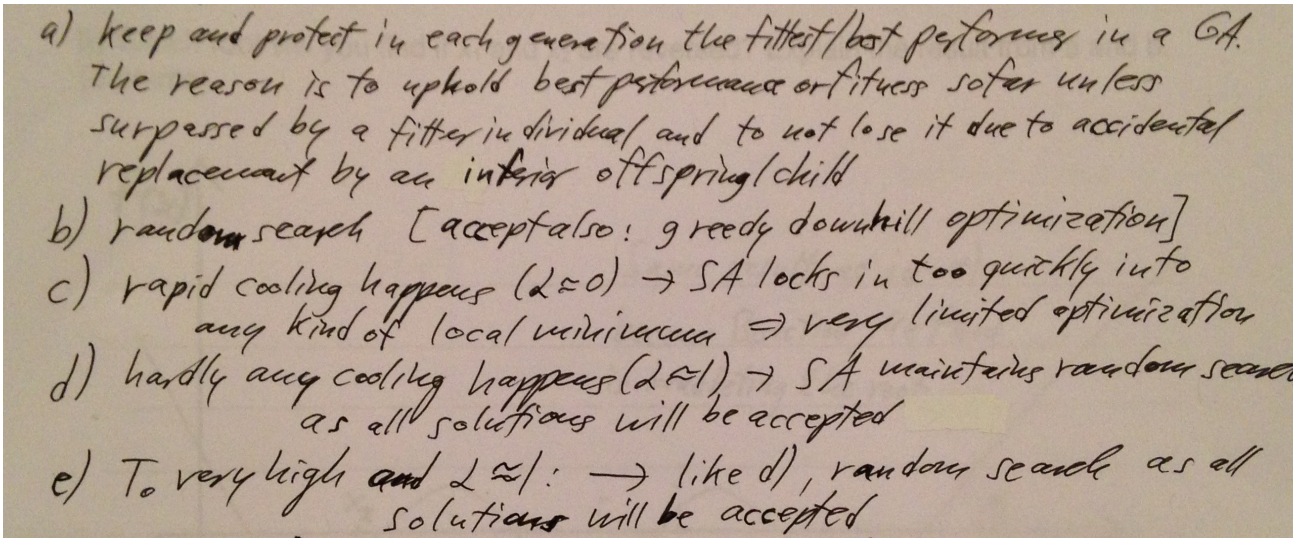
a) Explain the concept of the Elitist approach in Genetic Algorithms. Why is it important?
(2 points)

b) What does Simulated Annealing turn into when you eliminate the section of the algorithm that accepts a worse performing solution via the Boltzmann-probability?
(2 point)

c) What happens to Simulated Annealing when the cooling rate Lambda is very close to 0?
(2 points)

d) What happens to Simulated Annealing when the cooling rate Lambda is very close to 1?
(2 points)

e) What happens to Simulated Annealing when the initial temperature is very high and the cooling rate Lambda very close to 1?
(2 points)



Question #12 (16 points total):

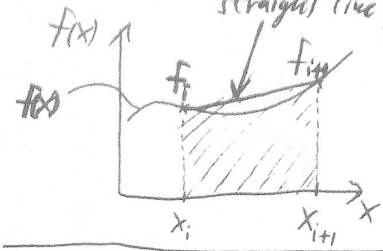
a) Graphically describe the Trapezoidal Rule and explain it. List the mathematical approximation formula for a single interval.
(4 points)

b) Graphically describe the Simpson Rule and explain it. List the mathematical approximation formula for a single interval.
(4 points)

c) Write down mathematically the Gauss-Legendre Integration (GLI) approximation of an integral of a function f(x) over an arbitrary finite interval [a,b]. Explain at what locations the function f(x) is being evaluated in the GLI approximation.
(5 points)

d) Explain for which types of integrands GLI is exact and why.
(3 points)

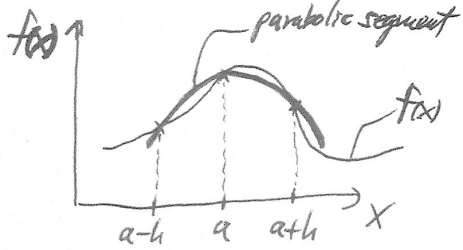
a) Trapezoidal Rule (TR):



TR approximates integral of function f(x) between each pair of neighboring points x_i, x_{i+1} by the area under the connecting straight line.

$$\int_{x_i}^{x_{i+1}} f(x)dx \approx (x_{i+1}-x_i)\,\frac{f(x_{i+1})+f(x_i)}{2}$$

b) Simpson Rule (SR):



SR constructs a parabolic arc through each triad of points x_i, x_{i+1}, x_{i+2} and integrates over the areas underneath the respective parabolic arcs.

$$\int_{a-h}^{a+h} f(x)dx \approx h\left[\frac{f(a-h)}{3}+\frac{4}{3}f(a)+\frac{f(a+h)}{3}\right]$$

c)

$$\int_a^b f(u)du = \frac{b-a}{2}\int_{-1}^{1} f\left(\frac{(b-a)x}{2}+\frac{b+a}{2}\right)dx$$

$$\text{GLI} \approx \frac{b-a}{2}\sum_{i=1}^{n} w_i f\left(\frac{(b-a)x_i}{2}+\frac{b+a}{2}\right)$$

with $x_{i=1,\dots n}$ = n zeros of Legendre pol. L_n in $]-1,1[$ ← degree
and w_i = integration weights

d) All polynomials up to degree $(2n-1)$ are integrated exactly over $[-1,1]$ because they can be exactly expanded in Legendre polynomials according to: $P_{n-1} = \sum_{i=0}^{n-1} a_i L_i$